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Teaching a robot what a toy is

Domain-specific toy object detection for child–robot interaction: developing and evaluating a YOLO11-based perception system for WordMate

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Ground-truth bridge annotations in a multi-object construction scene (Figure 4.2d).

01 The problem

A robot has to recognise the toy in front of a child before it can name it, ask about it, or respond to it. Most robot-assisted language-learning studies avoid this problem — they use tablets, scripted activities or a hidden human operator.

Object detection research, meanwhile, has been built around driving, surveillance and industry: large, upright, well-lit objects. It has not been built around a carpet covered in half-buried plastic bricks.

So: can a current detector learn a specific vocabulary of toys well enough, and fast enough, to hold up in a live interaction?

02 The dataset

- 1,176 RGB images of the real WordMate play objects, captured for this project
- GoPro HERO12 Black and Intel RealSense D435i
- 43 classes — barbecue and kitchen items, a doctor's set, animal figures, blocks, cups
- Single objects and cluttered play scenes; viewpoint, distance, lighting and occlusion all varied



Barbecue set, animal figures and building blocks — photographed in the room the robot works in.

aim

Aim

Build and evaluate a YOLO11-based toy detector that can support object-grounded interaction in educational social robotics.

Research questions

- How well does YOLO11 detect toys in scenes typical of child–robot interaction?
- Which classes are least reliable, and why do they fail?
- Is it fast enough for real-time use?

03 How it was built

01 CAPTURE → Two cameras, controlled and naturalistic play scenes

02 ANNOTATE → 44 classes boxed by hand in Roboflow

03 AUDIT → Model-assisted label review on held-out data

04 CONSOLIDATE → Taxonomy checked against CDI vocabulary norms

05 TRAIN → YOLO11s, COCO weights, 5-fold stratified CV

annotation_audit

Bad labels cost more than a bad architecture

Of 187 confirmed annotation errors, 84% were objects simply left unlabelled. Retraining on the corrected labels, scored against the same corrected validation set:

	PRECISION	RECALL	mAP@50	mAP@50:95
Original labels	0.858	0.780	0.856	0.595
Corrected labels	0.873	0.846	0.898	0.621
Change	**+0.015**	**+0.066**	**+0.042**	**+0.026**

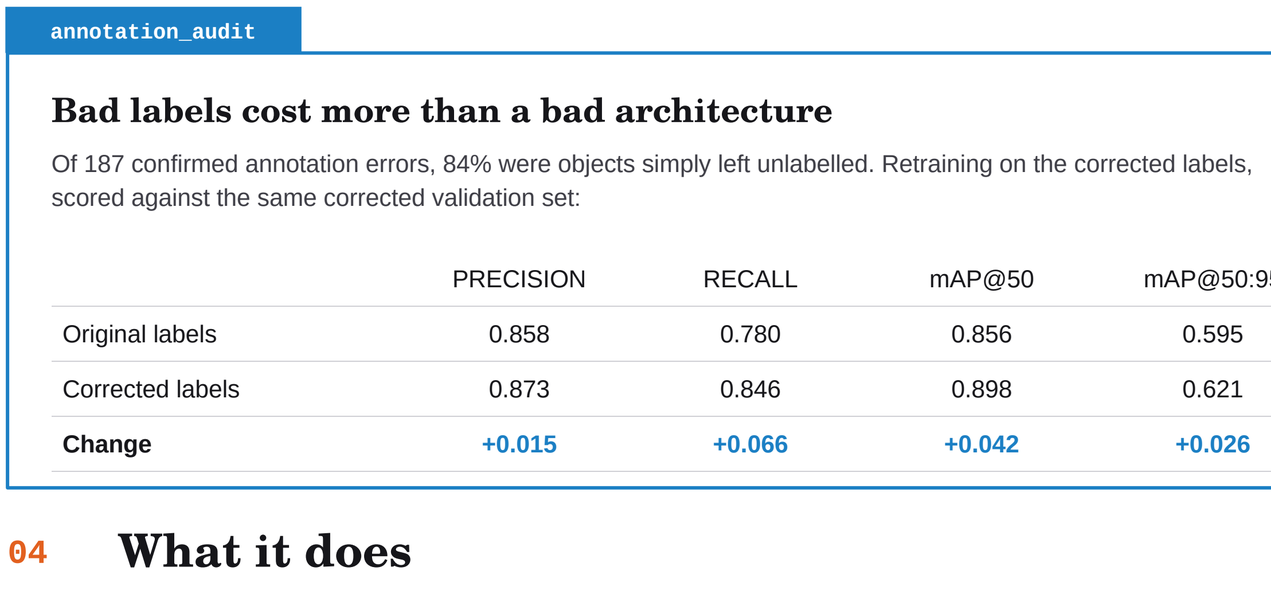
Training setup

- YOLO11s, COCO-pretrained, 640 × 640 input
- Five-fold iterative multi-label stratified cross-validation — every image held out exactly once, all 43 classes present in every fold
- Final-epoch checkpoint; no early stopping and no best-checkpoint selection, so nothing is tuned on the held-out data

04 What it does

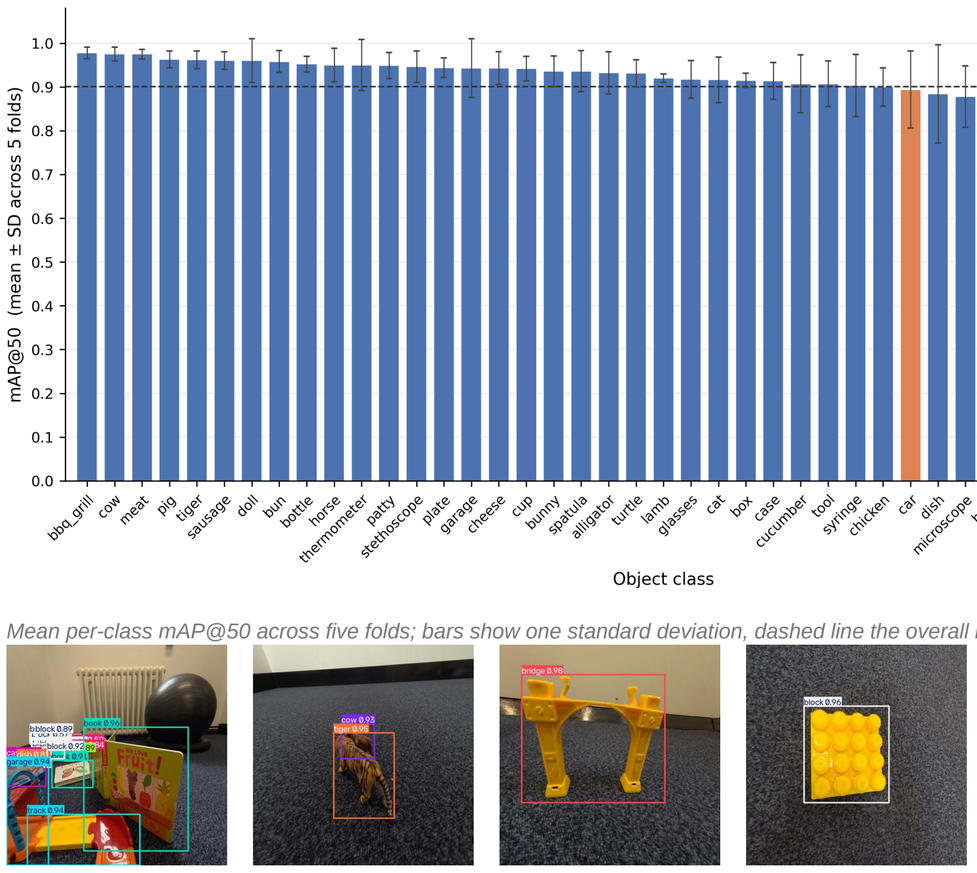
0.901	0.641	0.897	0.851	310.5
mAP@50 ± 0.011	**mAP@50:95** ± 0.015	**PRECISION** ± 0.014	**RECALL** ± 0.018	**FPS ON GPU** 24.7 on CPU

Mean ± SD across five held-out folds, 43 classes. Speed measured at 640 × 640, batch 1.



Mean per-class mAP@50 across five folds; bars show one standard deviation, dashed line the overall mean.

- Median class scored 0.920; 30 of the 43 classes reached 0.90 or better
- The two stragglers are track (0.513) and bridge (0.568) — long, thin, half-buried in the structures they belong to, and punished hard by strict overlap thresholds
- Failures were missed objects, not mistaken ones: no pair of classes was confused above 10%, so the 43-class vocabulary is visually coherent
- Adding a person class reached 0.990 mAP@50 with no observed cost to the toys
- Run live on camera for eight minutes it returned 13.7 objects per frame at confidence 0.25, and 9.4 at 0.50



Real output, not ground truth

Predictions from the bundled 44-class checkpoint on held-out validation images at confidence 0.25. Illustrative examples rather than a separate evaluation.

conclusion

Where this leaves WordMate

Accurate and fast enough to build on — but this is component-level feasibility, not a working classroom system. Next: annotated live-camera evaluation on the robot itself, targeted capture of occluded track and bridge pieces, and confidence thresholds tuned to the cost of a missed toy in conversation.

Ethics: Abertay University EMS11209 · Belpaeme et al. (2018) · van den Berghe et al. (2019) · Redmon et al. (2016) · Khanam & Hussain (2024) · Lin et al. (2014)